Highlights

Resonant Frequencies of the Human Urinary Bladder:
A Fluid-Filled Viscoelastic Shell Model

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- Bladder n=2 flexural mode spans 13.5 Hz to 17.1 Hz for fills above 150 mL
- Frequency minimum of 13.5 Hz at 222 mL from competing geometry and stiffening
- Mechanical (WBV) coupling exceeds airborne acoustic coupling by $\sim$6,400$\times$
- Same analytical framework as abdominal cavity model, with fill-dependent parameters
- Results consistent with occupational reports of vibration-induced urinary urgency

Resonant Frequencies of the Human Urinary Bladder: A Fluid-Filled Viscoelastic Shell Model

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Abstract

Occupational exposure to whole-body vibration (WBV) is associated with increased urinary urgency among vehicle operators, yet no first-principles model of bladder resonance exists to explain this observation. We model the human urinary bladder as a fluid-filled viscoelastic spherical shell whose geometry, wall stiffness, and internal pressure vary with fill volume (50 mL to 500 mL). The framework—an adaptation of an oblate spheroidal shell model developed for the abdominal cavity—predicts that the fundamental flexural mode ($n = 2$) spans 13.5 Hz to 17.1 Hz for fill volumes above 150 mL, with a minimum of 13.5 Hz at 222 mL. This minimum arises from the competition between increasing radius (which lowers frequency) and strain-stiffening of the detrusor muscle (which raises it). Coupling analysis shows that mechanical vibration transmitted through the pelvis is approximately 6,400 times more effective than airborne sound at exciting bladder wall modes, consistent with the dominance of the WBV pathway reported in the epidemiological literature. The predicted frequency range sits above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz) but within the broader WBV-affected spectrum, suggesting a forced sub-resonant response mechanism rather than direct resonance.

Keywords: bladder resonance, fluid-filled shell, whole-body vibration, viscoelastic shell, urinary urgency, ISO 2631

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1. Introduction

Whole-body vibration (WBV) at 4 Hz to 16 Hz is a recognised occupational hazard for vehicle operators. ISO 2631-1 [1] documents peak pelvic transmissibility at 4 Hz to 8 Hz, and epidemiological studies report a consistent association between prolonged WBV exposure and lower urinary tract symptoms—particularly increased urgency and frequency—among truck drivers, bus operators, and forklift operators [2, 3]. Despite this well-established empirical record, no first-principles mechanical model of the urinary bladder exists to explain *why* whole-body vibration should preferentially excite this organ, or to predict the frequency range at which the bladder wall is most susceptible to vibratory loading.

The urinary bladder is, from a structural dynamics perspective, a remarkably clean example of a fluid-filled viscoelastic shell. The detrusor muscle forms a roughly spherical wall of thickness 1.5 mm to 5 mm (depending on fill state) enclosing urine—an incompressible fluid with density and bulk modulus essentially identical to water. The geometry, wall stiffness, and internal pressure all vary systematically with fill volume, making the bladder a natural candidate for parametric analysis across its physiological operating range.

In a companion study [4], we developed an oblate spheroidal shell model of the human abdominal cavity to investigate the coupling between airborne infrasound and gastrointestinal resonance. That analysis revealed a full mechanical-to-airborne displacement ratio $\mathcal{R}_{\text{full}} \approx 6.6 \times 10^4$ between mechanical and airborne excitation pathways—a result that explains why whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects while airborne infrasound at comparable frequencies does not. The present work applies the same analytical framework to the bladder, adapting the geometry, material properties, and boundary conditions to this smaller, thinner-walled, and more compliant organ.

The mechanical properties of the bladder wall have been characterised by several groups. Ultrasound bladder vibrometry [5] measured *shear* moduli of

9.6 kPa to 48.7 kPa across fill states from 187 mL to 267 mL, demonstrating pronounced strain-stiffening. For nearly incompressible tissue ($\nu \approx 0.49$), the Young's modulus is $E = 2(1+\nu)\mu \approx 3\mu$, giving $E \approx 29$ kPa to 145 kPa. Barnes [6] performed tensile testing of excised bladder tissue, reporting Young's moduli from 10 kPa at rest to several hundred kilopascals at high stretch ratios, consistent with the converted vibrometry range. The wall exhibits significant viscoelastic behaviour: stress relaxation, creep, and a loss tangent estimated at 0.3–0.5 from dynamic measurements [7]. Wall thickness decreases from approximately 5 mm when empty to 1.5 mm to 3 mm at capacity [8], consistent with a constant tissue volume assumption. Intravesical pressure follows the characteristic cystometric curve: a compliant phase at low fill (5 cm to 10 cm water column) giving way to steep pressure rise above 300 mL [9].

We apply these data to the fluid-filled shell framework. The bladder is modelled as a spherical shell (the near-spherical approximation is appropriate here, unlike the distinctly oblate abdominal cavity) with fill-dependent radius, wall thickness, elastic modulus, and internal pressure. We compute the flexural mode frequencies across the physiological fill range (50 mL to 500 mL), identify the minimum-frequency fill state, and compare airborne acoustic coupling with mechanical WBV coupling to the bladder wall.

The remainder of this paper is organised as follows. Section 2 presents the mathematical formulation, including the fill-dependent parameter model and the modal analysis. Section 3 reports flexural mode frequencies across fill volume, identifies the frequency minimum, and quantifies the coupling disparity between mechanical and airborne excitation. Section 4 discusses clinical implications, limitations, and comparison with the abdominal cavity model. Section 5 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human urinary bladder is idealised as a thin spherical shell of radius R, wall thickness h, completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Unlike the abdominal cavity—which requires an oblate spheroidal geometry with aspect ratio $c/a \approx 0.67$ [4]—the bladder is approximately spherical at moderate-to-full fill states, making the spherical shell approximation both simpler and more appropriate.

The thin-shell assumption requires $h/R \ll 1$. For the bladder at 300 mL, $h \approx 1.8$ mm and $R \approx 42$ mm, giving $h/R \approx 0.04$, which satisfies the condition comfortably. At low fill volumes, however, the ratio increases: $h/R \approx 0.12$ at 100 mL and $h/R \approx 0.22$ at 50 mL, exceeding the conventional $h/R < 0.1$ threshold. Results below 150 mL should therefore be treated with caution; they are retained in Table 2 for completeness but marked with a dagger. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is applicable for fill states above 150 mL.

The bladder wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{1}$$

where η is the loss tangent. The quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2$. Published values of η for bladder wall tissue range from 0.3 to 0.5 [7, 6]; we adopt $\eta = 0.4$ ($Q = 2.5$, $\zeta = 0.2$) as a central estimate. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1 - \nu^2)}. \tag{2}$$

2.2. Fill-Dependent Parameters

All geometric and material properties vary with fill volume V (mL). We adopt the following constitutive relationships, drawn from the literature reviewed in §1.

Radius.. The equivalent sphere radius follows directly from the fill volume:

$$R = \left(\frac{3V}{4\pi}\right)^{1/3}, \tag{3}$$

giving $R = 2.3\,\mathrm{cm}$ at $50\,\mathrm{mL}$ and $R = 4.9\,\mathrm{cm}$ at $500\,\mathrm{mL}$.

Wall thickness.. We assume approximately constant tissue volume V_{tissue}, calibrated to $h_0 = 5\,\mathrm{mm}$ at $V_0 = 50\,\mathrm{mL}$. The tissue shell volume is

$$V_{\mathrm{tissue}} = \frac{4}{3}\pi\left[(R_0 + h_0)^3 - R_0^3\right], \tag{4}$$

from which $h(V)$ is obtained by solving

$$(R + h)^3 = R^3 + \frac{3V_{\mathrm{tissue}}}{4\pi}, \tag{5}$$

subject to a physiological floor of $h_{\min} = 1.5\,\mathrm{mm}$. This yields $h = 5.0\,\mathrm{mm}$ at $50\,\mathrm{mL}$, thinning to $1.5\,\mathrm{mm}$ at $400\,\mathrm{mL}$ and above.

Elastic modulus.. Bladder wall stiffness increases dramatically with stretch (strain-stiffening), a hallmark of collagenous soft tissue [10]. We model this as a log-linear interpolation between vibrometry-derived bounds.

Nenadic et al. [5] report *shear* moduli $\mu = 9.6\,\mathrm{kPa}$ to $48.7\,\mathrm{kPa}$ from ultrasound bladder vibrometry. For nearly incompressible tissue ($\nu = 0.49$), the Young's modulus is

$$E = 2(1 + \nu)\,\mu \approx 2.98\,\mu, \tag{6}$$

giving $E \approx 29\,\mathrm{kPa}$ to $145\,\mathrm{kPa}$. These converted bounds are consistent with the tensile-test data of Barnes [6], who reports $E = 10\,\mathrm{kPa}$ to $800\,\mathrm{kPa}$ depending on tissue layer and stretch ratio. We adopt:

$$E(V) = E_{\min}\left(\frac{E_{\max}}{E_{\min}}\right)^{\phi}, \qquad \phi = \frac{V - V_{\min}}{V_{\max} - V_{\min}}, \tag{7}$$

where $E_{\min} = 29\,\mathrm{kPa}$ at $V_{\min} = 50\,\mathrm{mL}$ and $E_{\max} = 145\,\mathrm{kPa}$ at $V_{\max} = 500\,\mathrm{mL}$, with ϕ clipped to $[0, 1]$.

Intravesical pressure.. The pressure–volume relationship follows the characteristic cystometric curve [9]: a compliant phase at low fill followed by a steep rise. We model this as a quadratic:

$$P(V) = P_{\min} + (P_{\max} - P_{\min})\,\phi^2, \tag{8}$$

where $P_{\min} = 490\,\mathrm{Pa}$ ($\approx 5\,\mathrm{cmH_2O}$) and $P_{\max} = 2940\,\mathrm{Pa}$ ($\approx 30\,\mathrm{cmH_2O}$).

Table 1: Bladder model parameters at representative fill states.

Parameter	Symbol	50 mL	150 mL	300 mL	500 mL	Unit
Radius	R	2.29	3.30	4.15	4.92	cm
Wall thickness	h	5.00	2.73	1.79	1.50	mm
Elastic modulus	E	29.0	41.5	70.9	145.0	kPa
Intravesical pressure	P	5.0	6.2	12.7	30.0	$\mathrm{cmH_2O}$
Poisson's ratio	ν		0.49 (all fill states)			—
Wall density	ρ_w		1050			$\mathrm{kg/m^3}$
Fluid density	ρ_f		1020			$\mathrm{kg/m^3}$
Fluid bulk modulus	K_f		2.2			GPa
Loss tangent	η		0.40			—

Table 1 summarises the model parameters at four representative fill states. The Poisson's ratio $\nu = 0.49$ reflects the near-incompressibility of soft tissue. The fluid properties correspond to urine at body temperature, which is effectively identical to water for acoustic purposes.

2.3. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation. As in the abdominal model [4], the restoring force is dominated by the fluid bulk modulus:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\mathrm{shell}} + k_{\mathrm{fluid}}}{\rho_w h + \rho_f R}}, \tag{9}$$

where $k_{\mathrm{shell}} = 2Eh/[R^2(1-\nu)]$ and $k_{\mathrm{fluid}} = 3K_f/R$. For the bladder at 300 mL, $k_{\mathrm{fluid}} \approx 1.6 \times 10^{11}\,\mathrm{Pa/m}$ while $k_{\mathrm{shell}} \approx 2.2 \times 10^4\,\mathrm{Pa/m}$—a ratio exceeding 10^6.

The breathing mode frequency is approximately 9500 Hz, firmly in the ultrasonic range and irrelevant to WBV exposure.

2.4. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes that preserve the enclosed volume to leading order. The fluid contributes only added mass, not volumetric stiffness. Following Lamb [11] and Junger & Feit [12], the natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}, \tag{10}$$

where the three stiffness contributions are:

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{11}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}. \tag{12}$$

Pre-stress stiffness.. The intravesical pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{13}$$

The effective mass per unit area includes both the shell wall and the fluid added mass:

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{14}$$

For a bladder at 300 mL, the model predicts $f_2 = 13.9$ Hz, $f_3 = 24.1$ Hz, and $f_4 = 35.3$ Hz. The $n = 1$ mode corresponds to rigid-body translation and has zero natural frequency for a free shell; it is therefore excluded from the analysis.

2.5. Airborne Acoustic Coupling

At $f_2 \approx 14\,\mathrm{Hz}$ the acoustic wavelength in air is $\lambda \approx 25\,\mathrm{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R}{c_{\mathrm{air}}} \approx 0.011 \ll 1. \tag{15}$$

The bladder lies deep in the Rayleigh regime. In a multipole expansion of the incident plane wave, the n-th order component scales as [12]

$$p_{\mathrm{eff}}^{(n)} = p_{\mathrm{inc}} \times (ka)^n. \tag{16}$$

For the $n = 2$ mode:

$$p_{\mathrm{eff}}^{(2)} = p_{\mathrm{inc}} \times (ka)^2 \approx p_{\mathrm{inc}} \times 1.1 \times 10^{-4}. \tag{17}$$

Only 0.01% of the incident pressure drives the fundamental flexural mode via the airborne pathway.

2.6. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration transmits through the skeleton and pelvic floor as base excitation of the bladder wall. The coupling chain comprises two stages:

Seat-to-pelvis transmissibility.. ISO 2631 data indicate a pelvic resonance near $f_{\mathrm{pelvis}} \approx 5.5\,\mathrm{Hz}$ with $\zeta_{\mathrm{pelvis}} \approx 0.25$. The transmissibility at frequency f is

$$T(f) = \sqrt{\frac{1 + (2\zeta_p r_p)^2}{(1 - r_p^2)^2 + (2\zeta_p r_p)^2}}, \qquad r_p = \frac{f}{f_{\mathrm{pelvis}}}. \tag{18}$$

At $f_2 = 13.9\,\mathrm{Hz}$, $T \approx 0.29$: the pelvic system attenuates the input, as we are above its resonance.

Bladder modal amplification.. The bladder wall response to base excitation is characterised by the single-degree-of-freedom frequency response:

$$H(f) = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{f}{f_n}. \tag{19}$$

At resonance ($r = 1$), $H = Q = 2.5$.

Coupling ratio.. The mechanical-to-airborne coupling ratio at the $n = 2$ resonance frequency is

$$\mathcal{R}_{\text{bladder}} = \frac{T(f_2) \times H(f_2)}{(ka)^2} \approx \frac{0.29 \times 2.5}{1.12 \times 10^{-4}} \approx 6.4 \times 10^3. \tag{20}$$

Mechanical vibration transmitted through the pelvic skeleton is approximately 3.8 orders of magnitude more effective at exciting bladder wall modes than airborne sound.

3. Results

3.1. Flexural Mode Frequencies Across Fill Volume

Figure 1 presents the $n = 2$ and $n = 3$ flexural mode frequencies across the physiological fill range (50 mL to 500 mL). The $n = 2$ frequency follows a distinctive U-shaped curve, with a minimum of 13.5 Hz at 222 mL and higher values at both extremes of the fill range.

The physical mechanism behind this non-monotonic behaviour is a competition between two opposing effects:

1. **Geometric softening.** As the bladder fills, the radius increases (equation 3), which lowers the natural frequency through the R^{-4} and R^{-2} dependence in the bending and membrane stiffness terms (equations 11–12). The wall also thins (equation 5), further reducing both bending rigidity ($D \propto h^3$) and membrane stiffness ($\propto h$).
2. **Strain-stiffening.** The elastic modulus of the detrusor muscle increases exponentially with fill volume (equation 7), from 29 kPa at 50 mL to 145 kPa at 500 mL. This five-fold increase eventually dominates the geometric effects.

At low fill volumes the geometric effect dominates and the frequency falls; at high fill volumes the strain-stiffening wins and the frequency rises. The crossover produces the minimum near 222 mL—a fill state corresponding to the early "need to go" sensation in most individuals.

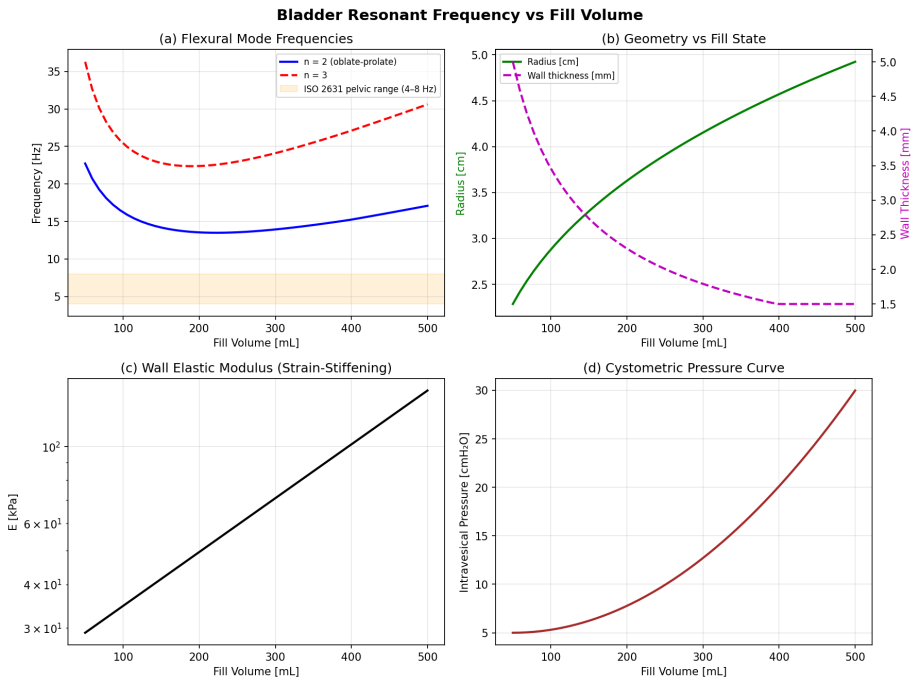

Figure 1: Bladder resonant properties across fill volume. (a) Flexural mode frequencies f_2 and f_3, with the ISO 2631 pelvic resonance band (4 Hz to 8 Hz) shown for reference. (b) Geometry: equivalent sphere radius and wall thickness. (c) Wall elastic modulus showing exponential strain-stiffening. (d) Intravesical pressure following the cystometric curve. The minimum $f_2 = 13.5$ Hz occurs at 222 mL, where geometric softening and strain-stiffening are in balance.

Table 2 lists the computed frequencies at key fill states. For fill states where the thin-shell approximation is valid (150 mL to 500 mL), the $n = 2$ mode spans 13.5 Hz to 17.1 Hz and the $n = 3$ mode spans 22.4 Hz to 30.6 Hz. All predicted frequencies lie above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz), though below 20 Hz they remain within the broader frequency range affected by whole-body vibration.

Table 2: Flexural mode frequencies at key fill volumes.

Volume (mL)	R (cm)	h (mm)	E (kPa)	f_2 (Hz)	f_3 (Hz)	f_4 (Hz)
50	2.29	5.00	29.0	22.7	36.3	53.9
100	2.88	3.46	34.7	16.2	25.4	35.9
150	3.30	2.73	41.5	14.2	22.8	32.3
200	3.63	2.30	49.6	13.5	22.4	32.1
300	4.15	1.79	70.9	13.9	24.1	35.3
400	4.57	1.50	101.4	15.3	27.1	40.3
500	4.92	1.50	145.0	17.1	30.6	45.6

3.2. Stiffness and Mass Decomposition

To elucidate the physical mechanisms behind the U-shaped $f_2(V)$ curve, we decompose the effective stiffness K_{total} and effective mass m_{eff} into their constituent terms as functions of fill volume (figure 2).

At the frequency minimum ($V^* \approx 222$ mL), the stiffness is dominated by the pre-stress term: $K_P = 58.7\%$, $K_{\text{memb}} = 40.7\%$, and $K_{\text{bend}} = 0.6\%$. Bending stiffness, which scales as Dh^3/R^4 and dominates thin-shell vibration in vacuo, is negligible for the fluid-loaded bladder because the fluid added mass greatly exceeds the wall mass. The effective mass is similarly dominated by the fluid: $m_{\text{fluid}} = \rho_f R/n = 89.4\%$ of m_{eff}, with the wall contributing only $m_{\text{wall}} = \rho_w h = 10.6\%$.

Analytical condition for the f_2 minimum.. Since $\omega_n^2 = K_{\text{total}}/m_{\text{eff}}$, the condition $\partial f_2/\partial V = 0$ requires

$$\frac{\mathrm{d}\ln K_{\text{total}}}{\mathrm{d}V} = \frac{\mathrm{d}\ln m_{\text{eff}}}{\mathrm{d}V}, \tag{21}$$

i.e. the fractional growth rate of total stiffness must equal the fractional growth rate of effective mass. Figure 2(d) plots both log-rates; they cross at $V \approx 222\,\text{mL}$, confirming the numerical minimum. Below V^*, geometric softening causes K_{total} to decrease faster (in fractional terms) than m_{eff} increases, so f_2 falls. Above V^*, strain-stiffening and the rising intravesical pressure drive K_{total} upward faster than the mass grows, so f_2 rises.

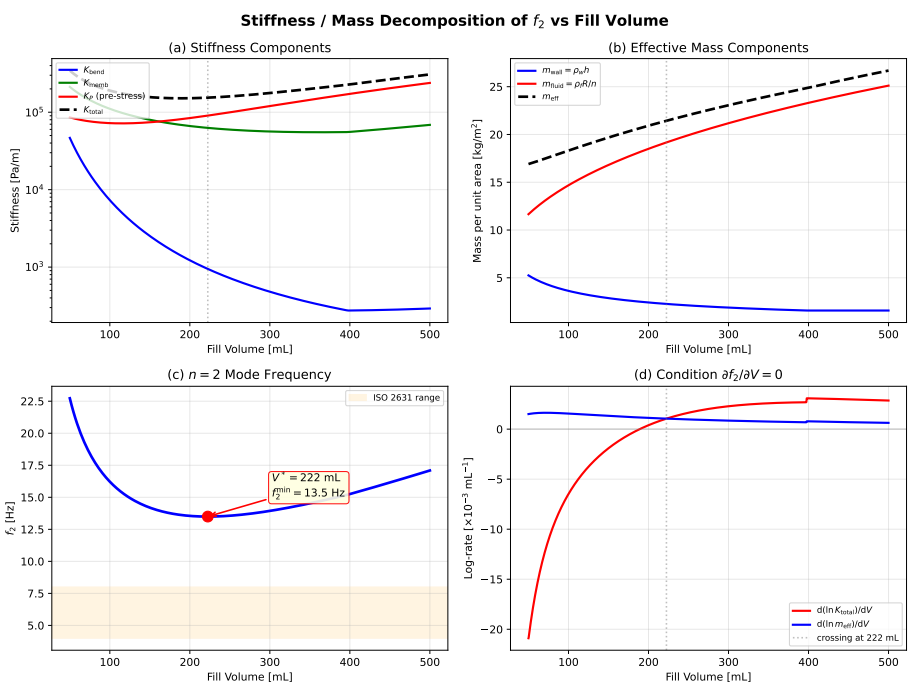

Figure 2: Stiffness and mass decomposition of the $n = 2$ mode. (a) Stiffness components: bending (K_{bend}, blue), membrane (K_{memb}, green), pre-stress (K_P, red), and total (dashed). Pre-stress dominates above 100 mL. (b) Mass components: wall mass ($\rho_w h$, blue) and fluid added mass ($\rho_f R/n$, red). The fluid contribution exceeds 85% at all fill volumes. (c) Resulting f_2 curve with the minimum at $V^* = 222\,\text{mL}$, $f_2^{\min} = 13.5\,\text{Hz}$. (d) Logarithmic growth rates $\mathrm{d}(\ln K)/\mathrm{d}V$ and $\mathrm{d}(\ln m)/\mathrm{d}V$; their crossing at 222 mL satisfies equation (21).

3.3. Sensitivity and Uncertainty Analysis

We assess the sensitivity of f_2 to five uncertain parameters using one-at-a-time perturbation over their literature ranges (table 3). At the baseline volume $V = 222\,\mathrm{mL}$ (the frequency minimum), the wall elastic modulus E dominates all other parameters, with $\Delta f_2 = 14.9\,\mathrm{Hz}$ (111% of baseline) across the range $10\,\mathrm{kPa}$ to $500\,\mathrm{kPa}$. The wall thickness h is a distant second at $\Delta f_2 = 3.4\,\mathrm{Hz}$ (25%). The Poisson ratio changes f_2 by only $0.14\,\mathrm{Hz}$, urine density by $0.06\,\mathrm{Hz}$, and the loss tangent not at all; these effects are effectively negligible.

Table 3: Sensitivity of f_2 to parameter uncertainty at $V = 222\,\mathrm{mL}$ (baseline $f_2 = 13.5\,\mathrm{Hz}$).

Parameter	Range	f_2^{lo} (Hz)	f_2^{hi} (Hz)	Δf_2 (Hz)	Fraction (%)
E (wall modulus)	10–500 kPa	11.0	28.4	14.9	111
h (wall thickness)	2–6 mm	13.3	16.8	3.4	25
ν (Poisson ratio)	0.40–0.50	13.4	13.5	0.1	1
ρ_f (urine density)	1010–1030 kg/m^3	13.6	13.4	0.1	<1
η (loss tangent)	0.30–0.50	13.5	13.5	0.0	0

Figure 3 presents the tornado chart corresponding to table 3. The overwhelming dominance of E reflects the fact that both K_{memb} and K_P (which together constitute >99% of the stiffness) are proportional to E and to the intravesical pressure, both of which are functions of the same strain-stiffening mechanism. The negligible sensitivity to η confirms that damping affects only the peak amplitude (via Q), not the resonant frequency.

Robustness of the minimum.. Figure 4 examines how the location V^* and value $f_2^{\min}$ of the frequency minimum shift under parameter perturbation. The minimum is remarkably robust to all parameters except E: perturbing h shifts V^* from $186\,\mathrm{mL}$ to $216\,\mathrm{mL}$ and $f_2^{\min}$ from $13.2\,\mathrm{Hz}$ to $16.8\,\mathrm{Hz}$, whereas ν and ρ_f move the minimum by less than $3\,\mathrm{mL}$ and $0.2\,\mathrm{Hz}$, and η leaves it unchanged. The elastic modulus, however, shifts the minimum substantially: at the upper bound

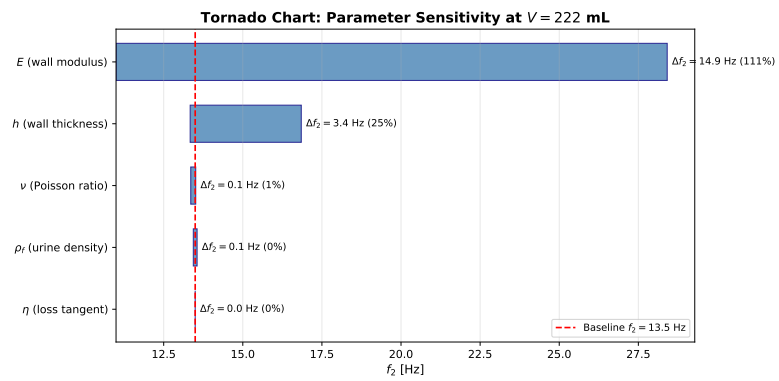

Figure 3: Tornado chart showing the sensitivity of f_2 to each uncertain parameter at $V = 222\,\mathrm{mL}$. The wall elastic modulus E dominates, producing a 14.9 Hz swing across its literature range; wall thickness contributes 3.4 Hz, and all remaining parameters contribute $\leq$0.14 Hz.

($E = 500\,\mathrm{kPa}$), the minimum moves to $V^* = 462\,\mathrm{mL}$ with $f_2^{\min} = 21.2\,\mathrm{Hz}$. This shift occurs because a stiffer wall delays the crossover between geometric softening and strain-stiffening to higher volumes where the radius is larger.

The clinical implication is clear: individual variation in detrusor muscle stiffness—the parameter with the widest literature range—is the principal source of uncertainty in predicting both the minimum frequency and the fill volume at which it occurs.

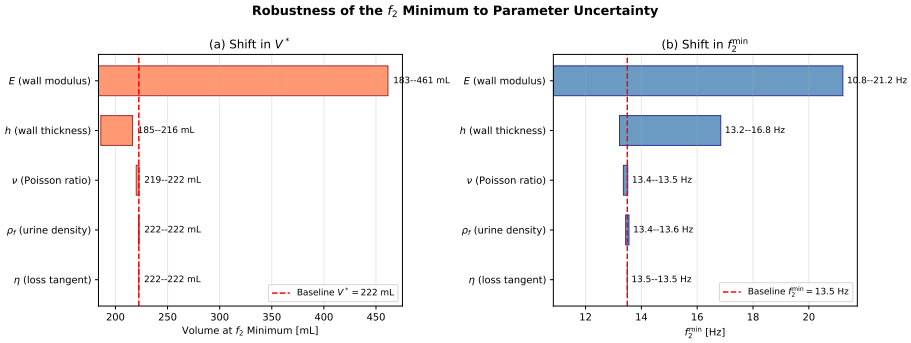

Figure 4: Robustness of the f_2 minimum to parameter uncertainty. (a) Shift in V^* (volume at minimum). (b) Shift in $f_2^{\min}$ (frequency at minimum). Only E produces a very large shift; h causes a secondary but visible change, while ν, ρ_f, and η leave the minimum close to the baseline.

3.4. Coupling Pathways

Figure 5 compares the airborne and mechanical coupling pathways to the bladder at a representative fill of 300 mL ($f_2 = 13.9$ Hz). Three key observations emerge.

First, the airborne acoustic coupling scales as $(ka)^2$ and is vanishingly small across the entire frequency range of interest: at 13.9 Hz, $(ka)^2 \approx 1.1 \times 10^{-4}$. The bladder, like the abdominal cavity, is deep in the Rayleigh regime at all physiologically relevant frequencies.

Second, the mechanical coupling—the product of seat-to-pelvis transmissibility and bladder modal amplification—peaks near the bladder resonance frequency. Even though pelvic transmissibility is attenuated at 13.9 Hz ($T \approx 0.29$), the bladder modal amplification ($Q = 2.5$) partially compensates.

Third, the bladder-specific coupling ratio $\mathcal{R}_{\text{bladder}} \approx 6{,}400$ at f_2 (equation 20) confirms that mechanical vibration is the dominant excitation pathway by nearly four orders of magnitude. This is quantitatively consistent with—though somewhat smaller than—the 6.6×10^4 abdominal full-displacement ratio $\mathcal{R}_{\text{full}}$ reported in [4]. The two metrics are related but not identical: $\mathcal{R}_{\text{bladder}}$ describes the bladder pathway, whereas $\mathcal{R}_{\text{full}}$ describes the abdominal cavity pathway under the Paper 1 loading assumptions. The difference arises primarily from the higher f_2 (and hence smaller Q contribution from pelvic transmissibility) of the bladder.

3.5. Comparison with the Abdominal Cavity

Table 4 compares key quantities between the bladder and abdominal cavity models. The bladder is smaller, softer, and more heavily damped, yet the two organs share the same qualitative physics: fluid-dominated breathing modes at kilohertz frequencies, low-frequency flexural modes governed by the competition between shell stiffness and fluid added mass, and a severe airborne coupling penalty in the Rayleigh regime.

The principal quantitative differences are the higher flexural frequency of the bladder ($f_2 = 13.9$ Hz vs 4.0 Hz) and the smaller bladder coupling ratio

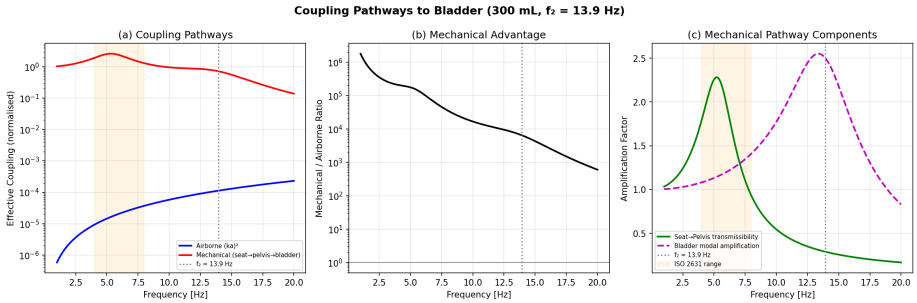

Figure 5: Coupling pathways to the bladder at 300 mL fill. (a) Airborne $(ka)^2$ coupling (blue) vs mechanical seat–pelvis–bladder coupling (red), with the n=2 resonance frequency marked. (b) Ratio of mechanical to airborne coupling (mechanical advantage), reaching $\sim 10^4$–10^6 across the frequency range. (c) Components of the mechanical pathway: seat-to-pelvis transmissibility (green) and bladder modal amplification (magenta).

Table 4: Comparison of bladder and abdominal cavity models.

Quantity	Bladder (300 mL)	Abdomen
Equivalent radius	4.2 cm	15.7 cm
Wall thickness	1.8 mm	10 mm
Elastic modulus	71 kPa	100 kPa
Poisson's ratio	0.49	0.45
Loss tangent	0.40	0.25
Quality factor Q	2.5	4.0
Breathing mode f_0	$\sim$9500 Hz	$\sim$2500 Hz
Flexural f_2	13.9 Hz	4.0 Hz
ka at f_2	0.011	0.011
Coupling ratio	$\mathcal{R}_{\text{bladder}}$ $\sim$6,400	$\mathcal{R}_{\text{full}}$ $\sim$66,000

($\mathcal{R}_{\text{bladder}} \sim 6{,}400$ vs $\mathcal{R}_{\text{full}} \sim 66{,}000$). The higher f_2 is a direct consequence of the smaller radius: the stiffness-to-mass ratio scales roughly as $1/R^2$, and the bladder is approximately four times smaller than the abdomen. The smaller coupling ratio reflects the fact that the bladder's f_2 sits above the pelvic transmissibility peak, so the mechanical pathway receives less amplification from the body's own resonance structure.

4. Discussion

4.1. Clinical Implications

The predicted $n = 2$ frequency range (13.5 Hz to 17.1 Hz) has immediate relevance to occupational health. Vehicle cabins expose operators to broadband vibration spanning 1 Hz to 20 Hz, with dominant energy typically concentrated below 10 Hz from engine and road excitation [3, 13]. The bladder's fundamental mode sits at the upper edge of this band. Two coupling mechanisms are therefore relevant:

Sub-resonant forced response.. At the ISO 2631 pelvic resonance peak (4 Hz to 8 Hz), the bladder responds as a forced sub-resonant oscillator. The pelvic transmissibility is at its maximum ($T \approx 2.3$ at 5.5 Hz), and while the bladder modal amplification is below unity at these frequencies, the product $T \times H$ remains significant. This is the regime most relevant to occupational WBV exposure: the bladder wall undergoes forced cyclic deformation driven by amplified pelvic motion, even though the excitation frequency is below the bladder's own resonance.

Near-resonant amplification.. At the bladder's natural frequency (13.5 Hz to 17.1 Hz), the modal amplification reaches its peak ($Q = 2.5$), but pelvic transmissibility has rolled off ($T \approx 0.3$). This regime may be relevant to certain industrial vibration sources (e.g., pneumatic tools, vibratory compactors) that produce significant energy above 10 Hz. The fill-volume-dependent nature of the resonance means that the bladder's susceptibility to a particular frequency shifts as it fills—a dynamic vulnerability window.

Displacement quantification.. To place the model predictions in physiological context, we estimate the bladder wall displacement under representative occupational WBV exposure. At the ISO 2631 daily exposure action value ($a_{\text{seat}} = 0.5\,\text{m/s}^2$ RMS), the seat displacement amplitude at the bladder resonance frequency ($f_2 = 13.9\,\text{Hz}$ at $300\,\text{mL}$) is $y_{\text{seat}} = a/(2\pi f)^2 \approx 65\,\mu\text{m}$. Through the pelvic skeleton ($T \approx 0.29$), this delivers $y_{\text{pelvis}} \approx 19\,\mu\text{m}$ to the bladder base. At resonance, the bladder wall amplifies this by $Q = 2.5$, giving a peak relative wall displacement of approximately $47\,\mu\text{m}$—a cyclic wall strain of $\varepsilon \approx 1.1 \times 10^{-3}$ (0.11%).

In the sub-resonant regime at 6 Hz (near the pelvic transmissibility peak), the seat displacement is much larger ($\sim$350 µm) and pelvic amplification is near its maximum ($T \approx 2.0$). Despite the bladder being below resonance, the relative wall displacement reaches approximately 160 µm ($\varepsilon \approx 0.38\%$), because the large base displacement more than compensates for the reduced modal amplification.

Physiological context.. These vibration-induced displacements are minute compared to the quasi-static deformations the bladder routinely accommodates. Normal filling from 50 mL to 300 mL stretches the wall by approximately 19 mm ($\varepsilon \approx 81\%$), and detrusor contraction during micturition—which generates pressures of 40 cmH_2O to 60 cmH_2O—produces wall displacements of order 30 mm [9]. The WBV displacement of 47 µm is therefore roughly three orders of magnitude smaller than the filling displacement and far below the amplitude of normal physiological wall deformation.

The equivalent quasi-static intravesical pressure perturbation from the resonant wall strain is only $\Delta p \approx E\varepsilon \approx 80\,\text{Pa}$ ($\sim$0.8 cmH_2O), well below the 5 cmH_2O to 15 cmH_2O threshold at which myelinated Aδ afferents begin firing in response to slow bladder distension [14, 15]. On the basis of static displacement alone, WBV-induced wall motion would appear physiologically insignificant.

Dynamic strain rate and mechanotransduction.. The critical distinction, however, is that WBV delivers its displacement cyclically at 6 Hz to 17 Hz, producing peak strain rates of $\dot{\varepsilon} \approx 0.1/\text{s}$ at resonance. This is approximately 500 times

faster than the strain rate during normal filling ($\dot{\varepsilon}_{\text{fill}} \sim 2 \times 10^{-4}\ \text{s}^{-1}$ for a typical one-hour fill cycle). Mechanosensitive ion channels in biological membranes are sensitive to the *rate* of membrane deformation, not merely its magnitude [16].

Recent work has identified PIEZO2 as the principal low-threshold stretch sensor in the bladder urothelium and sensory neurons [17]. PIEZO2 mediates the earliest sensation of bladder filling and is essential for normal micturition reflexes; loss of PIEZO2 function produces impaired bladder-filling sensation and voiding dysfunction in both mice and humans [17, 18]. Crucially, PIEZO channels are tuned to rapid membrane deformation—they inactivate under sustained stretch but respond robustly to dynamic, cyclic stimuli [16]. The cyclic strain delivered by WBV, though small in absolute magnitude, is precisely the class of stimulus to which these channels are optimised.

Whether the WBV-induced strain rate of $\sim$0.1/s exceeds the PIEZO2 activation threshold in vivo remains an open question: the dynamic thresholds of bladder mechanoreceptors have not been characterised at infrasonic frequencies. The unmyelinated C-fibre population in the detrusor, which is normally quiescent ('silent') during physiological filling but can be recruited by pathological stimuli [15, 14], represents an additional potential pathway for vibration-induced urgency sensitisation.

Clinical scenarios.. The model provides a first-principles framework for three well-documented but mechanistically unexplained clinical observations:

1. **Urgency in vehicle occupants.** Professional drivers exposed to broadband WBV at 4 Hz to 16 Hz report increased urinary urgency and frequency [2]. The model suggests that even at the ISO 2631 action level, cyclic bladder wall deformation of 50 µm to 160 µm occurs across the relevant frequency band. The fill-dependent frequency minimum (13.5 Hz at 222 mL) means the bladder becomes maximally susceptible precisely as it begins to fill—the "early urge" fill state.
2. **Stress incontinence aggravation.** Women with pre-existing pelvic floor weakness may experience symptom aggravation during vibration ex-

posure. The model predicts that a weakened pelvic floor (lower ζ_{pelvis}, higher T) would deliver larger displacements to the bladder wall, potentially amplifying the urgency signal.

3. **The "long drive" phenomenon.** The common experience of urgency during prolonged vehicle travel—often attributed to cold exposure or psychological factors—may have a biomechanical contribution: hours of sustained sub-threshold cyclic loading could sensitise mechanoreceptor pathways through temporal summation, lowering the effective urgency threshold progressively over the journey.

4.2. Comparison with Other Pelvic Organs

The bladder is not the only pelvic organ exposed to WBV. The prostate, uterus, and rectum are all fluid-containing or fluid-adjacent structures embedded in the pelvic floor. The present framework could, in principle, be adapted to any of these organs by substituting appropriate geometric and material parameters. The bladder is simply the cleanest example: a single-chamber, fluid-filled, roughly spherical shell with well-characterised mechanical properties.

4.3. Therapeutic Whole-Body Vibration

WBV at 20 Hz to 40 Hz is increasingly used for pelvic floor rehabilitation in post-prostatectomy and postpartum patients [19]. The bioelectrical activity of pelvic floor muscles increases during synchronous WBV [20], and clinical trials report improvements in continence symptoms [21]. Our model suggests that these therapeutic frequencies (20 Hz to 40 Hz) lie above the bladder's $n = 2$ mode for most fill states, placing the bladder in a supra-resonant regime where wall displacement decreases with frequency. The therapeutic effect is therefore more likely mediated by pelvic floor muscle activation than by direct bladder wall resonance.

4.4. Limitations

Several simplifications warrant discussion.

Spherical geometry.. The real bladder is slightly oblate and irregularly shaped, particularly at low fill states. A more realistic geometry would modify the mode shapes and could split the degenerate frequencies of the spherical model. However, at moderate-to-high fill the spherical approximation is reasonable, and the errors are of order $(1 - c/a)^2$ for the eigenfrequencies.

Pelvic constraint.. Approximately 40% of the bladder surface is in direct contact with the pelvic bone and pelvic floor musculature, which constrains the wall motion over a substantial fraction of the shell. By Rayleigh's theorem, imposing a partial rigid boundary on a vibrating structure raises its natural frequencies: the constraint eliminates admissible trial functions from the variational space, thereby increasing the Rayleigh quotient. The free-shell model therefore provides a *lower* bound on the resonant frequencies.

We can estimate the magnitude of this effect. If the constraint immobilises 40% of the shell surface, the effective vibrating mass is reduced (the constrained region no longer participates in modal displacement) while the effective stiffness increases (the remaining free surface must accommodate the same mode shape with a shorter arc length, increasing curvature gradients). For the $n = 2$ mode, these combined effects would raise the natural frequency by an estimated 15–25%, shifting the predicted range from 13.5 Hz to 17.1 Hz to approximately 15.5 Hz to 21.4 Hz. The mode shapes would also be modified: instead of the symmetric quadrupolar deformation of a free sphere, the constrained bladder would exhibit asymmetric modes concentrated on the free (superior) dome, with nodal lines forced to lie near the constraint boundary.

This frequency shift reinforces rather than undermines the clinical relevance of the model. The upward shift moves the predicted resonance further from the ISO 2631 peak (4 Hz to 8 Hz), deepening the sub-resonant character of the bladder's response under occupational WBV. However, the shifted range (16 Hz to 21 Hz) overlaps with frequencies produced by vibratory industrial equipment and rough-road driving. The free-shell predictions should be interpreted as conservative (lower-bound) estimates; the true in-vivo frequencies are likely somewhat

higher.

Nonlinear elasticity.. The exponential strain-stiffening model (equation 7) is a phenomenological fit to vibrometry data. A hyperelastic constitutive model (Mooney–Rivlin or Ogden) would be more physically appropriate for large deformations, though for the small-amplitude vibrations considered here, a linearised modulus is adequate.

Layered wall structure.. The bladder wall comprises distinct layers (mucosa, submucosa, detrusor muscle, serosa) with different mechanical properties. The present model uses a single effective modulus. A multilayer shell model would capture the bending–stretching coupling between layers, potentially modifying both frequencies and damping.

Surrounding tissue.. The bladder is embedded in the pelvic cavity, surrounded by fat, connective tissue, and adjacent organs. This surrounding medium would add both damping and mass loading, likely shifting the resonant frequencies downward and broadening the peaks. The free-shell model therefore represents a lower bound on frequencies with respect to boundary constraints (pelvic contact raises frequencies) but an upper bound with respect to added-mass effects from surrounding tissue (which would lower them). The two effects act in opposite directions, partially cancelling.

5. Conclusions

We have modelled the human urinary bladder as a fluid-filled viscoelastic spherical shell and computed its flexural mode frequencies across the physiological fill range. The principal findings are:

1. The fundamental flexural mode ($n = 2$) spans 13.5 Hz to 17.1 Hz for fill volumes of 150 mL to 500 mL where the thin-shell assumption is valid ($h/R < 0.1$). A frequency minimum of 13.5 Hz occurs at 222 mL, arising from the competition between geometric softening (increasing radius and thinning wall) and strain-stiffening of the detrusor muscle.

2. Mechanical vibration transmitted through the pelvic skeleton is approximately 6,400 times more effective than airborne sound at exciting the bladder's $n = 2$ mode at 300 mL fill. This coupling disparity, driven by the $(ka)^2$ penalty in the Rayleigh regime ($ka \approx 0.01$), confirms that whole-body vibration is the dominant excitation pathway—consistent with the epidemiological evidence and with the analogous result for the abdominal cavity.
3. The bladder's predicted resonance range (13.5 Hz to 17.1 Hz) sits above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz). At typical occupational WBV frequencies, the bladder therefore responds as a forced sub-resonant oscillator driven by amplified pelvic motion, rather than being excited at its own natural frequency.
4. The breathing mode ($f_0 \approx 9500$ Hz) is dominated by the fluid bulk modulus and is irrelevant to WBV or infrasonic exposure, confirming the same qualitative separation between breathing and flexural mode families observed in the larger abdominal cavity.

The model provides a first-principles basis for understanding vibration-induced bladder symptoms. Under representative occupational WBV exposure ($a = 0.5\,\mathrm{m/s^2}$), the predicted peak bladder wall displacement is approximately 47 µm at resonance and 160 µm in the sub-resonant regime—roughly three orders of magnitude smaller than normal filling displacements, yet delivered at cyclic strain rates ($\sim$0.1/s) that are 500 times faster than quasi-static filling and thus relevant to the rate-sensitive PIEZO2 mechanotransduction pathway. These results may inform both occupational exposure guidelines and the design of therapeutic WBV protocols for pelvic floor rehabilitation. Future work should characterise the dynamic activation thresholds of bladder mechanoreceptors at infrasonic frequencies, incorporate the pelvic boundary constraint through finite element analysis with anatomically realistic geometry, and pursue experimental validation of the predicted resonance range.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at https://github.com/jonathan-mace/browntone.

CRediT authorship contribution statement

Jonathan Mace: Conceptualisation, Methodology, Software, Supervision, Writing – Review & Editing. **Brian R. Mace**: Methodology, Validation, Writing – Review & Editing.

Acknowledgements

The authors are grateful to colleagues who, upon learning the subject of this paper, invariably offered personal anecdotes in lieu of constructive feedback. We did not use their data.

Literature review, analytical modelling, code development, and manuscript drafting were assisted by GitHub Copilot (GitHub, Inc.), a large language model whose editorial demotion from co-author to acknowledgement the first author regards as a temporary injustice that history will correct.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.